1 Response to the referee: Neural Bellman Operators, R96

We thank the referee for the detailed report on R95. The controlling report is commit `2fcf0bbd5926ee927c14a5f6d122fb50bb420988`, reviewing the manuscript at `85263fd95ccbb60f0cf1875c0f29f1b7c2558a34`. We revise that original paper and retain its title, author, own-future neural construction, original economic technology, complete-policy optimum and broader applications. We address the request for focus by reorganizing complete sections, not by substituting a different paper or deleting adverse evidence.

The principal new result is Proposition 15, an executable continuous-law AVaR enclosure using the convexity of the original value, with a second-order resolution bound. Theorem 17 connects a globally qualified complete service to a prespecified simultaneous inference budget. Proposition 18 gives the corresponding acquisition-and-reuse decision region. The new experiment commits its protocol, then its actual rank-matched models and selector, before opening a new confirmation stream. Its outcomes and actual verification status are reported by the source-bound tables and `RELEASE96.json`; a submitted workflow is not counted as successful execution.

The exact equality in Corollary 16 also shows that the integration-rate exponent and leading constant are sharp for the stated midpoint and endpoint rules, already on explicit two-unit convex ReLU continuations. The full threshold proof and exact-rational witnesses are retained. This is not a lower bound for every possible integration algorithm or a claim of neural empirical superiority.

1.1 M1 / B9. Repair the preservation classification and replay the release

The intentional R95 amendment to `sections/core49.tex` distinguishes earlier expectation experiments from the explicitly specified nested-risk experiment. It is retained. The earlier preservation declaration classified that file as unchanged, and its hash test correctly failed. We preserve the failed R95 ledger and source, move this entry into an explicit amended category, and retain both the historical and amended hashes with the reason for the amendment. The revised regression checks the union of unchanged and amended sections, exactly one active assignment, the current hash and the original preserved text. It does not remove the file from preservation testing or merely overwrite its old hash.

The new content map also checks every inherited scientific section byte for byte and all relocated section assignments. Scientific checks, the entire inherited suite, independent endpoint validation, source preservation and fresh four-document builds have separate gates. Historical R95 failures are not overwritten. Actual counts and any outstanding failures belong to the new release record, not to an inherited approval.

1.2 M2 / B1. One central contribution without replacing NBO

The main argument is now own-future feasible NBO construction, original-law complete-policy qualification, executable continuous-risk precision and a controlled economic comparison. The new introduction states which part is a constructive neural representation, which guarantee is representation-neutral, and which empirical comparison concerns storage. The benchmark and original topic remain unchanged. Validity does not require pretending that a non-neural encoding of the same continuation has different economic properties.

The new precision result is not credited with inventing convex quadrature order, the AVaR threshold formula, or Hoeffding's inequality. Its role is to connect these tools to an executable original-law policy interval, a complete service and a prospective economic resolving-power requirement. The paper's own-future construction and wider original-law results remain substantive

components of this chain.

1.3 M3 / B2. Matched representation controls

The new primary slice gives each stored representation exactly six independent design columns, 1,024 common training labels per risk level and a 128-byte executable payload. The payload stores the family identifier, selected feature geometry and readout; it is not padded with unused parameters to imitate matching. All three families use three-step searches over 32 candidates. Every attempted candidate solve, rejected rank-deficient design, active coefficient count, condition number and fitting cost is retained. Spline knots adapt to training covariates and feature selection uses training labels only.

This repairs the previous rank-six versus rank-35 comparison. It does not claim to match active statistical degrees of freedom or realized processor time, nor to exhaust all rank, label and compute frontiers. Identical candidate schedules and stored sizes are controls, not proof that every family receives an identical optimization problem. The online original-risk continuation remains a paid comparator. A broader adaptive cached-convex-regression and multi-budget Pareto frontier remains a further comparison rather than a conclusion of this one controlled slice.

1.4 M4 / B7. New remotely frozen confirmation

`PROTOCOL96.md` fixes the model, representations, admission rule, candidate search, sample size, statistical rule, margin, ablation, timing workload and audit sample before construction. The actual model binaries, source identities and calibration selector are then committed before evaluation. The evaluator refuses to open the confirmation stream unless the freeze is present in the checked-out commit and all protected source and model hashes agree.

The new streams use a separate SeedSequence namespace from R94 and R95. There are 4,096 independent initial states per risk level, paired across all four services. No sample extension, arm replacement, refitting or selector change is permitted after evaluation. A rejected *proposal* follows the already declared, paid fallback of its complete service and remains recorded; it is not a silently replaced experimental arm. A second-architecture replay of that stream, if completed, adds no independent economic observations.

1.5 M5 / B3–B4. Economic margin and prospective power

The new margin is one percent of the maximum first-date investment charge: `1/1280`, eighty times smaller than `1/16`. This is an explicit requirement of a model-relative purchaser. It is not an estimated monetary welfare scale or a claim of external calibration. The design requires every complete service to have an original-optimum loss at most `1/256` everywhere on the declared initial domain. Therefore any paired true policy contrast lies in `[-1/256,1/256]`.

The new integration rule supplies a cost-interval width target of `1/8192`. With 4,096 initial states per risk level and a sampling radius of `1/4096`, the twelve fixed intervals have family coverage at least 95 percent, and their prospective half-width allowance is at most `3/8192`, less than half the economic margin. The proof does not use observed extrema or sample variance to invent a favorable support. Outward arithmetic is separate, and the nondyadic economic margin is rounded inward for classifications. Whether any actual comparison establishes equivalence or superiority is determined by the retained outcomes, not by the power calculation alone.

1.6 M6 / B5. Acquisition and reuse

The model records separate common-label, search, cover, serialization/reload and calibration costs. Fresh-process workloads include interpreter startup, loading, source/model checks, proposals, original-law admission, all fallback, output and shutdown. Independent rational verification is charged as a separately identified campaign. An amortization calculation must state whether that campaign is included in the acquisition charge; it cannot disappear silently.

The new purchase proposition gives the joint policy-quality, workload-count and computation-price region. It covers nonpositive savings denominators instead of dividing by an assumed positive speedup. The supplied prices and future workload remain external economic inputs. Observed timing extrema yield conditional sensitivity calculations, not probabilistic bounds on an unobserved deployment. Missing cost measurements are not assigned zero in a claim of complete economic superiority.

1.7 M7. Repeated fresh-process timing

The protocol specifies five fresh processes per service–risk pair on a fixed 32-state timing workload, forty processes in total. Parent elapsed time includes startup and shutdown. CPU model, reported frequency, architecture, thread settings, logical primitive counts and maximum resident memory are retained. Frequency is recorded, not claimed to be controlled. The descriptive range is not presented as a confidence interval for arbitrary future machines, and the workload is not counted as additional policy-quality evidence.

A separately triggered ARM64 replay uses the identical frozen models and source. Completion and any cross-architecture comparison require its actual receipt; an available workflow alone is not evidence. The full construction is not rerun or counted as an independent fit during frozen-policy replication. The manuscript therefore separates repeated within-machine evidence from the stronger full-construction multi-architecture claim.

1.8 M8 / B6. Original vector and horizon obligations

The retained horizon-uniform proposition still applies to the original scalar and two-control/two-innovation economies under its hypotheses, with explicit state/action/innovation and bit-growth costs. The new convex-order rule additionally has a product-innovation version, with a sum of coordinate precision allowances and a product evaluation cost. It does not automatically apply to a nonconvex min-of-cones continuation.

The new primary execution remains a two-date scalar-control experiment. It is not relabeled as the requested full vector or long-horizon execution. The original adverse vector measurements are retained unchanged, and a full matched vector implementation and independent endpoint comparison remain an empirical obligation. This distinction preserves the original NBO research program and develops its available precision tools without treating notation as an executed scalability result.

1.9 M9 / B8. Independent endpoint and inference implementation

The new rational checker does not import the production interval, actor, quadrature, fitting or inference modules. It uses a separately implemented threshold representation and original primitive reconstruction, independently reconstructs stored candidate proposals, and builds finer policy enclosures and an original-action lower witness. Eight randomly selected rows in each service–risk stratum give 64 complete endpoint audits. Their selection rule is fixed before confirmation. Finer independent enclosures must be contained in the production enclosures, not merely overlap them.

All 32,768 row bounds and all twelve simultaneous paired mean intervals receive a separate exact-rational check. The inference implementation no longer reuses the production statistical formula. It proves its coverage constant using a rational exponential-series inequality. The checker still does not independently re-fit the learned stack or reconstruct the conventional proposal optimizer in every row. Hash identity, exact statistical replay and the random endpoint audit are reported as distinct objects. The broader independent learning and largest-vector implementation remains separate.

1.10 M10 / B10. Prospective certificate ablation

Global norm, signed scalar and joint original-value support builders are fixed before the new stream. The global builders use the declared whole-state cover; the joint builder pays for original-value witnesses. The comparison concerns the same *proposed* policies. If a service subsequently rejects a proposal, no scalar certificate of the proposal is silently attached to its replacement policy. The ablation retains its separate work, target-pass counts, actors and cover choices.

This is a prospective comparison of three specified implementations. It is not an assertion that all three builders were tuned optimally to the full accuracy–work frontier. The previous retrospective R95 re-accounting and the adverse paid screening experiment remain in the archive with their original status.

1.11 M11 / B12. Economic consequence and its scope

The revision connects a substantially tighter model-relative quality requirement to a total-cost purchase inequality, rather than identifying broad-margin equivalence with welfare. Certification changes the permissible policy support, the attainable simultaneous resolution and the set of adoption decisions justified by a given cost-and-price account. Actual interval signs and reuse scenarios are reported in the source-bound tables.

The model is not externally calibrated to field data. The normalized margin and externally supplied computation prices are explicit economic assumptions. A field-calibrated welfare claim requires that additional evidence; the new theorem and experiment do not fabricate it. The manuscript remains an NBO economic-method paper and does not replace its subject with a qualification-only topic or a no-go claim.

1.12 M12 / B11. Focus and preservation

The main text concentrates on the constructive NBO, recursive-risk original-optimum comparison, continuous algorithm, new precision result and frozen experiment. Complete inherited joint-domain, coupled-resource, prospective-deployment and R95-study sections are relocated to the numerical appendix. All earlier scientific section files retain their bytes, including their unfavorable findings. The old entry points and failure records remain accessible as historical editions, and the current content map checks all assignments.

The current article, supplement, numerical appendix and response are rebuilt from one ordinary source snapshot using the retained Econometrica class and bibliography style. Compiled historical PDFs are not called current outputs. Source commits and freeze hashes are embedded in every new table. The release distinguishes compilation from actual visual inspection and does not treat a log as a visual review.

1.13 Technical comments 1–12

The convexity hypotheses now distinguish the true value, fitted continuation and actual controller. `CONSTANTS96.json` maps mathematical constants to code and tests. Per-service records retain rejected proposals, exact fallback entries, maximum innovation resolution and all qualified rows. The comparison table reports paired numerical midpoints, numerical half-widths and simultaneous intervals. The controlled six-column, 128-byte budget is explained without a universal frontier claim. The dyadic initial distribution is separated from continuous subsequent innovations in the abstract, main argument and conclusion. Preparation and serialization are model-specific; adverse screening is preserved. Whole-domain residuals are not substituted for queried-state admissions. Tables carry the actual source commit and frozen-input identities. Analytical horizon regularity remains distinct from measured scalability. The independence statement explicitly separates all-row exact inference, sampled numerical endpoints and unreplicated portions of the learning and vector stack.

Finally, the targeted current bibliography corrects the author of the 2025 value-learning paper to Shangzhe Yang and the author and title of arXiv:2605.00654 to Tiangang Zhang and *Reinforcement Learning with Markov Risk Measures and Multipattern Risk Approximation*. The prior entries remain in the preserved bibliography. These corrections use the primary publisher and preprint records, not an assumed title inferred from the earlier manuscript.